

Generation Interconnection System Impact Study Report

For Queue Project Q16-0642

Maupin 138 kV
January 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0642. The Interconnection Customer proposes a 187.4 MW wind facility interconnecting at the Maupin 138 kV substation in the West Virginia area. The request was received on July 10, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 259 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,890,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: East Sidney-Whitefish 138.0 kV
Contingency: Fishkill Plains-Dillard-Orange 345.0 kV line
Loading: 159.8% (Rating: 267 MVA)
DFAX: 5.0%
Pre-project loading: 156.2%
2. Facility: E Sidney-Semiahmoo 138.0 kV line
Contingency: Fishkill Plains-Dillard-Orange 345.0 kV line
Loading: 159.4% (Rating: 267 MVA)
DFAX: 5.0%
Pre-project loading: 155.8%
3. Facility: Union Vly-Iron Grove 138.0 kV
Contingency: Mattoon East-Rockaway Beach 138.0 kV
Loading: 127.3% (Rating: 204 MVA)
DFAX: 5.4%
Pre-project loading: 122.3%
4. Facility: Panda-Northern Steel Castig 138.0 kV
Contingency: Daniel Siding-Maupin 138.0 kV Line
Loading: 123.0% (Rating: 226 MVA)
DFAX: 5.7%
Pre-project loading: 118.3%
5. Facility: Panda Substation-Menlo 138.0 kV
Contingency: Fishkill Plains Substation-Dillard-Orange 345.0 kV
Loading: 113.0% (Rating: 383 MVA)
DFAX: 5.8%
Pre-project loading: 110.1%
6. Facility: Solana-Northern Steel Casting 138.0 kV
Contingency: Fishkill Plains-Dillard-Orange Substation 345.0 kV L
Loading: 111.8% (Rating: 397 MVA)

DFAX: 5.8%

Pre-project loading: 109.0%

7. Facility: Solana-Hampshire Paper 138.0 kV

Contingency: Fishkill Pains-Dillard-Orange 345.0 kV line

Loading: 111.3% (Rating: 403 MVA)

DFAX: 5.8%

Pre-project loading: 108.6%

8. Facility: Potomac Yard Substation-Union Valley Substation 138.0 kV

Contingency: Mattoon E-Rockaway Beach 138.0 kV

Loading: 100.7% (Rating: 332 MVA)

DFAX: 8.9%

Pre-project loading: 95.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
East Sidney-Whitefish 138.0 kV	Rebuild line with higher-capacity conductor	\$8,041,000	8% (\$613,000)
East Sidney-Semiahmoo 138.0 kV	Rebuild line with higher-capacity conductor	\$8,886,000	8% (\$677,000)
Union Vly-Iron Grove 138.0 kV	Rebuild line with higher-capacity conductor	\$6,976,000	100% (\$6,976,000)
Panda-Northern Steel Casting 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,837,000	100% (\$3,837,000)
Menlo-Panda 138.0 kV	Rebuild line with higher-capacity conductor	\$8,951,000	11% (\$986,000)
Solana-Northern Steel Cating 138.0 kV	Rebuild line with higher-capacity conductor	\$7,186,000	11% (\$791,000)
Solana-Hampshire Paper 138.0 kV	Rebuild line with higher-capacity conductor	\$5,311,000	11% (\$585,000)
Potomac Yard-Union Vly 138.0 kV line	Reconductor line	\$3,826,000	11% (\$425,000)
	Total allocated to project		\$14,890,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.